



华中科技大学 2024~2025 学年第一学期
“C 语言程序设计”考试试卷 (A 卷)

考试方式 闭卷 考试时间 2025 年 01 月 14 日上午 考试时长 150 分钟

题号	一	二	三	四	五	六	总分	总分人	核对人
分值	10	10	10	10	25	35	100		
得分									

得分	
评卷人	

一、判断下列语句或程序段的对错。(“×”表示错,“√”表示对)
(10 分) (Judge the correctness of the following statements or program segments. (“×” indicates wrong, and “√” indicates correct.))

- (1) struct AIA {
 char name[30] = "Li Hua";
 int age = 20;
} chen, gao, wang; (X) *定义结构时不能对成员初始化*
- (2) double *p=0xFFE0; (X) *只能写变量地址, 数组名不能初始化指针变量*
- (3) char *p_str = "Econ"; (√)
- (4) float bb[5], *p=bb;
 *p=10.0; (√)
- (5) int aa[10] = {0}; (√)
- (6) int (*px)[3], x[3][5];
 px = x; (X) *第二维大小不一致*
- (7) register int a;
 scanf("%d", &a); (X) *register 只能用在局部变量, 不能取址*
- (8) char str[30] = "AIA", *ptr = "ECON";
 strcat(str, ptr); (X) *标识符定义错误*
- (9) float xy#1 = 80.0; (X)
- (10) char *str = "\\abcd\\'efg\\'"; (√)

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二、计算下列表达式的值 (10 分) (Calculate the value of the following expressions.)

假设 unsigned int a=4, b=5; int c=3, d=6; float e=1.0;
char s1[20]="ECON", s2[20]='A','T','A','\0','H','U','S','T';

计算下列各表达式的值, 假设 int 类型为 16 位长度, 且各题彼此独立

(Suppose unsigned int a = 4, b = 5; int c = 3, d = 6; float e = 1.0;

char s1[20] = "ECON", s2[20] = {'A', 'T', 'A', '\0', 'H', 'U', 'S', 'T'};

Calculate the value of each of the following expressions, assuming that the int type is 16 bits long and each question is independent of each other.)

已算 改变d的值
--d
a非0, 不执行

- (1) $d \geq 1$, $a \parallel \neg d$, d ; (3 6)
 (2) $(int)(e/d + a/c) + (int)((e+b)/c)$; (3)
 (3) $c = strlen(strcat(s2, s1))$; 注意s2中间有'\0', 为"AJA" (7)
 (4) $a += (!b + a * c + 1)$; (17)
 (5) $b \&= 0x55AA$; (0)

$\Rightarrow 0x000580x55AA = 0$

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三、程序改错 (10 分) 要求: 不得改变程序框架, 不得重写程序, 无需文字说明, 直接在代码上添加、删除和修改。(Program Debugging: Do not change the program framework, do not rewrite the program, and make additions, deletions, and modifications directly on the code without written

explanations.)

(1) 如下程序对已知的 5 个整数从小到大排序, 其中在 main() 函数中初始化, 在 ssort 函数中排序。(The following program sorts 10 known integers in ascending order. The integers are initialized in the main() function and sorted in the ssort function.)

```
#include<stdio.h>
#define N 5
void ssort(char *pb, int m);
void main()
{
    int i;
    int b[N]={888,666,999,333,777};
    ssort(b,N);
    for(i=0; i<N; i++)
    {
        printf("%s\n", s[i]) j printf("%d\n", b[i]);
    }
}

void ssort(char *pb, int m)
{
    int i, j, k;
    char *t;
    for(i = 0; i < m; i++)
    {
        k = i;
        for(j = 0; j < m-1; j++) for(j=i+1; j<m; j++)
        {
            if(pb[k] > pb[j]) k = j;
        }
        if(k == i); k != i if(k != i)
        {
            t = pb[k];
            pb[k] = pb[i];
            pb[i] = t;
        }
    }
}
```

(2) 逐个比较 a,b 两个字符串中对应位置的字符, 把 a 字符串中 ASCII 值大或等于 b 字符串中对应位置字符依次存放到 c 数组中, 行成一个新的字符串, 并在主函数中输出 (Compare the characters at corresponding positions in the two strings a and b one by one, and store the characters in string a whose ASCII values are greater than or equal to the corresponding characters in string b into the c array in sequence to form a new string, and output it in the main function.)(5 分)

```
#include<stdio.h>
#include<string.h>
void fun(char *a, char *b, char *c);
void main()
{
    char a[10]="aBCDefgh",b[10]="ABcd",c[80]='\0';
    fun(a,b,c);
    puts(c);
}
void fun(char *a,char *b,char *c)
{
    int k = 1;  int k=0;
    while(*a != *b)  while (*a != '\0' || *b != '\0')
    {
        if(*a < *b)
            c[k] = *b;
        else
            c[k] = *a;
        if(*a)
            a++;
        if(*b)
            b++;
        k++;
    }
    c[k] = '\0';
}
```

得分	
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四、程序填空 (10 分)。

(1) x 从键盘输入, 计算 $\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots + (-1)^n \frac{x^{2n}}{(2n)!}$, 并使最后一项的绝对值小于 0.000001 为止, 最后输出 x 和计算结果。(x is input from the keyboard, $\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots + (-1)^n \frac{x^{2n}}{(2n)!}$ is calculated, and the calculation stops when the absolute value of the last term is less than 0.000001. Finally, output x and the calculation result.)

```

#include <stdio.h>
void main()
{
    int i = 1;
    double x;
    double an, bn, sum = 0;
    char s;
    printf("input x : \n");
    scanf("%lf", &x);
    s = 1;
    an = 1.0;
    bn = 1.0;
    while(an/bn >= 1e-6)
    {
        sum += s * an / bn;
        an = an * x * x;
        bn = bn * i * (i + 1);
        s *= -1;
        i++;
    }
    printf("cos(%f) = %f\n", x, sum);
}

```

(2) 输入一行字符串，统计其中数字和各个小写字母出现的次数。(Input a string and count the occurrences of numeric characters and each lowercase letter.)

```

#include <string.h>
#include <stdio.h>
void main()
{
    char str[200];
    int i, num, s_alpha[26];
    gets(str);
    num = 0;
    for(i=0; str[i] != '\0'; i++)
    {
        if( str[i] >= 'a' && str[i] <= 'z' )
            s_alpha[str[i] - 'a']++;
        else if( str[i] >= '0' && str[i] <= '9' )
            num++;
    }
    for(i=0; i<26; i++)
        printf("small letter %c : %d\n", 'a'+i, s_alpha[i]);
    printf("number : %d\n", num);
}

```

得分	
评卷人	

五、输出程序运行结果(Read the program and write the result.)(25 分)

```
(1)
#include <stdio.h>
void main()
{
    int i;
    for(i=2; i<10; i++)
    {
        if(i%4 == 0)
        {
            printf("to ");
        }
        else if(i%5 == 0)
        {
            break;
        }
        else
        {
            printf("Welcome ");
            continue;
        }
    }
    printf("HUST!");
}
```

Welcome Welcome to HUST!

```
(2)
#include <stdio.h>
int n;
void func();
void main()
{
    int i;
    for(i = 0; i < 5; i++)
    {
        func( );
    }
}
void func()
{
    int m = 2;
    static int j = 1;
    m *= 2;
    j *= (n + 1);
    printf("n=%d, m=%d, j=%d\n", ++n, m, j);
}
```

n=1, m=4, j=1
n=2, m=4, j=2
n=3, m=4, j=6
n=4, m=4, j=24
n=5, m=4, j=120

```
(3)
#include <stdio.h>
#define N 5
void main()
{
    int a[N][N];
    int i, j;
    for(i=0; i<N; i++)
```

```

{
    a[i][0]=1;
    a[i][i]=1;
}
for (i=2;i<N;i++)
{
    for (j=1;j<i;j++)
    {
        a[i][j]=a[i-1][j-1]+a[i-1][j];
    }
}
for(i=2;i<N;i++)
{
    for(j=0;j<=i;j++)
    {
        printf("%5d",a[i][j]);
    }
    printf("\n");
}
}

```

```

1   2   1
1   3   3   1
1   4   6   4   1

```

```

(4)
#include <stdio.h>
struct TEST
{
    char *str ;
    int num;
};
void main( )
{
    struct TEST a[ ] = {"AIA", 987}, {"HUST", 654}, {"CHINA", 321}, {"ASIA", 2025}};
    struct TEST *p ;
    int b=0, i=1;
    char *ss ;
    p = a + 3;
    ss = ++ (p->str) ;
    puts(ss);
    for(p=a; p< a + 4; p++)
    {
        b = b*10 + (p->num/i)%10;
        i *=10;
    }
    printf("b=%d\n", b) ;
}

```

SIA

b=7532

```

(5)
#include<stdio.h>
void main( )
{
    char *str[ ] = {"HUST","WUHAN","HUBEI","_CHINA"};
    char **p[ ] = {str+3, str+2, str+1, str};
    char ***ppp = p;
    printf("%s---", *(*++ppp));
    printf("%s", *(*ppp+1)+1);
}

```

HUBEI---CHINA

$*(*++ppp) \Rightarrow *(*(p+1)) \Rightarrow *(p[1]) \Rightarrow *(str+2) \Rightarrow str[2]$

$\Rightarrow ppp = p+1$ 第6页 共8页

$*(**ppp+1)+1 \Rightarrow *(*(p+1)+1) \Rightarrow *(str[p[1]+1])+1 \Rightarrow *(str+3)+1 \Rightarrow str[3]+1$
 指向c